Appendix: Sample data analysis and uncertainty propagation

All values below use the reported uncertainty convention: instrument half-resolution bounds are converted to standard uncertainties by dividing by $\sqrt{3}$, assuming uniform rounding errors. The uncertainty of locating fringes is assumed to be $\sigma_N = 0.5$ fringe. Common reference errors are not independent observations.

Green laser: wavelength from fitted slope

$$\begin{aligned} N &= m_\lambda \Delta x + b, \\ \lambda &= \frac{2000}{m_\lambda}\ \text{nm}, \\ \sigma_\lambda &= \left|\frac{\partial \lambda}{\partial m_\lambda}\right| \sigma_m = \frac{2000\sigma_m}{m_\lambda^2}. \end{aligned}$$

For green trial 1, $m_\lambda = 3.60253 \pm 0.07300$ fringes/μm, hence $\lambda_1 = 555.17 \pm 11.25$ nm.

Measurement uncertainty in mirror displacement

If each micrometer reading has a maximum rounding error of $0.5\,\mu$m, $\sigma_{x_i} = 0.5/\sqrt{3}\,\mu$m. For a difference $\Delta x_i = x_i - x_0$, the standard uncertainty of a single difference is

$$\sigma_{\Delta x} = \sqrt{\sigma_{x_i}^2 + \sigma_{x_0}^2} = \sqrt{2}\frac{0.5}{\sqrt{3}}\,\mu\text{m}.$$

The same x_0 is shared by the data, so the point-to-point covariance is not zero. A free intercept absorbs a constant reference offset when estimating the slope.

Angular transformation and refractive index

$$\begin{aligned} \theta &= \frac{\pi}{180}\theta_{\text{deg}}, \qquad u = \theta^2, \\ \sigma_u &= \left|\frac{du}{d\theta}\right| \sigma_\theta = 2|\theta|\sigma_\theta, \\ n &= \frac{1}{1 - m_n \lambda/t}. \end{aligned}$$

At $\theta = 1^\circ$, $u = 0.0003046$ rad^2 and with a single-angle standard uncertainty of 0.0050383 rad, $\sigma_u = 0.0001759$ rad^2.

For independent slope, wavelength, and thickness uncertainties,

$$\sigma_n = n^2\sqrt{\left(\frac{\lambda\sigma_m}{t}\right)^2 + \left(\frac{m\sigma_\lambda}{t}\right)^2 + \left(\frac{m\lambda\sigma_t}{t^2}\right)^2}.$$

For the 134°+ direction $m_n = 6678.853 \pm 272.118$ rad^{-2}, $t = 7790000$ nm, and $\lambda = 550.710$ nm. This gives $n = 1.8945$ and $\sigma_n = 0.0718$. The shared-reference-angle bias is an additional possible systematic error, treated separately.

Temperature difference and aluminium expansion

$$\begin{aligned}
\Delta T_i &= T_i - T_0, \\
\sigma_{\Delta T_i} &= \sqrt{\sigma_{T_i}^2 + \sigma_{T_0}^2}, \\
\alpha &= \frac{m_\alpha \lambda}{2L_0}, \\
\frac{\sigma_\alpha}{|\alpha|} &= \sqrt{\left(\frac{\sigma_{m_\alpha}}{m_\alpha}\right)^2 + \left(\frac{\sigma_\lambda}{\lambda}\right)^2 + \left(\frac{\sigma_{L_0}}{L_0}\right)^2}.
\end{aligned}$$

At $T = 27.0°\text{C}$, $T_0 = 25.2°\text{C}$ and each reading has $\sigma_T = 0.05/\sqrt{3}$ C, the difference is $\Delta T = 1.8°\text{C}$ with $\sigma_{\Delta T} = 0.0408°\text{C}$. The fitted slope is $m_\alpha = 6.95191 \pm 0.04032$ fringes/°C, giving $\alpha = (21.479 \pm 0.280) \times 10^{-6}\,°\text{C}^{-1}$.

Goodness of fit

$$\chi^2 = \sum_i \frac{(y_i - \widehat{y}_i)^2}{\sigma_{y_i}^2}, \qquad \chi^2_\nu = \frac{\chi^2}{N - p}$$

The above conventional expression applies to independent errors in y only. Our primary two-axis fit minimizes squared normalized corrections in both x and y, and reports the corresponding total χ^2 divided by $N - p$. $R^2 = 1 - \text{SSE/SST}$ is computed from vertical residuals at recorded x-values. Exact reference-angle and thermal-lag systematics are not captured by these reported statistical uncertainties.

Course handout: OLS slope, uncertainty, and residual

$$m = \frac{N\sum_i x_i y_i - (\sum_i x_i)(\sum_i y_i)}{N\sum_i x_i^2 - (\sum_i x_i)^2}, \qquad s_m = \sqrt{\frac{N s_{y,x}^2}{\Delta}}, \qquad s_{y,x}^2 = \frac{\sum_i (y_i - m x_i - b)^2}{N - 2}.$$

For green trial 1, $N = 5$ and $\Delta = 1250$. The ordinary least-squares slope is $m = 3.6000 \pm 0.0611$ fringes/μm. At $\Delta x = 5.0\,\mu$m, the observed fringe count is 15, the prediction is 14.200, and residual is 0.800 fringes. The effective vertical residual-bar uncertainty in a linear fit is $\sigma_{r,\text{eff}} \simeq \sqrt{\sigma_N^2 + (m\sigma_{x,\text{ind}})^2}$.